

She-Hulk: Lawyer to Fashion Icon

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Figure 1: During the development stage, the Digital Domain team captured She-Hulk on their motion capture stage in Los Angeles.

ABSTRACT

Digital Domain's work on She-Hulk: Attorney Law, which earned the visual effects (VFX) studio a Telly Awards win and Visual Effects Society Awards nomination, is heavily focused on the digital human work initially introduced in the "Avengers: Infinity War" for Thanos done by the same team four years earlier. To bring She-Hulk (played by Tatiana Maslany) to life, Digital Domain worked closely with the Marvel Studios' creative team to create a new type of superhero, a fashion icon and the best lawyer in town. Digital Domain's Visual Effects Supervisors, Jan Philip Cramer and Michael Melchiorre, led a team of 700 artists in four different locations, from Vancouver and Montreal to Los Angeles and Hyderabad. The show itself started in early September 2020 — in the middle of the pandemic — with final delivery in September 2022. With She-Hulk being our first show to be produced end-to-end from home, it meant that most of the team met only online in a work-from-home setup. Early on Digital Domain organized three tech days in our Los Angeles-based mocap studio to help define Maslany's (She-Hulk) motion and determine

the best shoot approach for this massive Disney+ visual effects show.

CCS CONCEPTS

• **Computing methodologies**; • **Machine learning**; • **Artificial intelligence**;

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1 MASQUERADE 2.0 & MACHINE LEARNING

A core element became the use of optical and inertial based motion capture for her body dynamics. Due to the many locations, inertial mocap was used on location, while larger sets were covered with optical motion capture that the Digital Domain Animation team would refine under the supervision of Elizabeth Bernard and Daniel Broverman. To capture Tatiana's (She-Hulk) facial performances, Techno Prop Helmet Cams (HMCs) were used.

Digital Domain uses machine learning algorithms to track the facial markers via our proprietary facial solver Masquerade 2.0. The result provides the animator with a much closer representation of She-Hulk actress, Tatiana's, performance and allows them to focus on the nuances of her performance. The facial rigs at Digital Domain use a FACS-based system (Facial Action Coding System),

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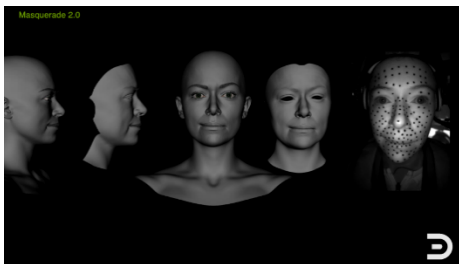


Figure 2: Masquerade 2.0 and facial solve quality control process showing the transformation from Tatiana Maslany to She-Hulk.



Figure 3: CFX simulation from skeleton to muscle to final skin.

so the animators still have detailed control over the character’s facial muscles and can alter or adjust their performances, if needed. The animation team would focus closely on matching Tatiana’s performance as much as possible while respecting She-Hulk’s larger proportion and making subtle adjustments to maintain her larger scale. As seen in the show, the storyline required Tatiana to show her inner struggle with her colleagues, dates and clients, so the animators needed to follow the subtle details of Tatiana’s performance closely to keep the subtext matching her intent.

2 CHARACTER EFFECTS

Once Animation was completed, the Character Effects (CFX) team would have their work cut out, with three different grooms, 12 outfits, and a full muscle system riding underneath for every shot. The costumes themselves had to match their life-action counterparts in look and simulation, while always appearing flattering on She-Hulk. Most of the CFX work was handled in Houdini, from muscles to cloth, while her hair used Digital Domain’s proprietary grooming tool, Samson. She-Hulk’s hair also needed to feel real but have a 70s-like volume, and with a special shader to allow for her subtle shimmering hair extensions and tips.

3 LOOK DEVELOPMENT

The look development (lookdev) for She-Hulk had a huge focus on makeup. As the typical computer-generated (CG) characters are monstrous villains, makeup became a crucial development for the team at Digital Domain. The use of contour makeup would help refine She-Hulk’s face shape. Lipstick and eye makeup also became a crucial aspect to her look. Initially, this was handled via maps



Figure 4: Use of Charlatan She-Hulk transformation.

in comp. Once approved, this look would be rolled back into the She-Hulk base shader to make future shots faster and easier.

4 CHARLATAN & AI

In addition to Masquerade 2.0, Digital Domain used Charlatan — a deep-fake like artificial intelligence (AI) approach for digital faces. This technique was used for all She-Hulk transformation shots as well as stunt face-replacements. Digital Domain captured training material of the actors with five simultaneously running cameras in different lighting conditions. This, in turn, provides the opportunity to replace the actor’s face with ease while maintaining the original performance. Especially when blending a CG face back into life action photography of Tatiana, this tool became very handy, as it provided the compers with an element to bridge the CG into the plate for a seamless transition. For Digital Domain, this was its first episodic to use AI face-replacement. The team overall feels this will be a major pillar in VFX work going forward, as most pipeline steps can be optimized with the use of AI; whether it is lookdev, roto, or animation, the options are endless to improve the quality and speed of VFX productions.